Instructor: Prof. Bartolomeo Stellato AI: Yanjun Liu

ORF522 Fall 2025 Homework 5

Due Thursday 12/4 at 11:59pm via Gradescope

1. *ADMM for smart grid device coordination.* We consider an electrical grid consisting of N devices that exchange electricity over T time periods. Device i has energy profile $p^i \in \mathbf{R}^T$, with p_t^i denoting the energy consumed by device i in time period t, for $t = 1, \ldots, T,\ i = 1, \ldots, N$. (When $p_t^i < 0$, device i is producing energy in time period t.)

   Each device has a convex objective function $f_i : \mathbf{R}^T \to \mathbf{R}$, which we also use to encode constraints, by setting $f_i(p^i) = \infty$ for profiles that violate the constraints of device i. In each time period the energy flow has to balance, which means

$$\sum_{i=1}^{N} p_t^i = 0, \quad t = 1, \ldots, T.$$

   The optimal profile coordination problem is to minimize the total cost, $\sum_{i=1}^{N} f_i(p^i)$, subject to the balance constraint, with variables $p^i, i = 1, \ldots, N$.

   In this problem you will use ADMM to solve the optimal profile coordination problem in a distributed way, with each device optimizing its own profile, and exchanging messages to coordinate all of the profiles.

   From this point on, we consider a specific (and small) problem instance. There are three devices: a generator, a fixed load, and a battery, with cost functions described below.

   - *Generator.* The generator has upper and lower generator limits:

$$p^{\min} \leq -p_t \leq p^{\max}, \quad t = 1, \ldots, T.$$

     Note the minus sign, since a generator's profile is typically negative, using our convention. The objective is

$$f_{\text{gen}}(p) = \sum_{t=1}^{T} (\alpha(-p_t)^2 + \beta(-p_t)),$$

     where $\alpha, \beta > 0$ are given constants.

   - *Fixed load.* The fixed load has zero objective function and the constraint that its power profile must equal a given consumption profile $d \in \mathbf{R}^T$.

- *Battery.* The battery has zero objective function, and charge/discharge limits given by C and D, respectively:

$$-D \leq p_t \leq C, \quad t = 1, \ldots, T.$$

The battery is initially uncharged (*i.e.,* $q_1 = 0$), so its charge level in period t is $q_t = \sum_{\tau=1}^{t-1} p_\tau$ (we neglect losses for this problem). The charge level must be nonnegative, and cannot exceed the battery capacity:

$$0 \leq q_t \leq Q, \quad t = 1, \ldots, T+1.$$

The charge level is extended to time $T+1$ to allow the battery to charge/discharge in time T, subject to the operational constraints.

(a) Use CVXPY to solve the problem with data generated by following codes.

```
import numpy as np
T = 100                    # planning horizon
alpha, beta = 0.2, 1       # generator cost parameters
pmin, pmax = 0, 10         # generator limits
Q, C, D = 5, 1, 1          # battery parameters

# fixed load schedule
d = np.zeros((T,1))
d[0:20]=3; d[20:56]=5; d[56:60]=11;
d[60:80]=4; d[80:90]=10; d[90:100]=3
```

Plot the (optimal) power profile for the generator and battery, as well as the battery charge level. Show that the optimal value is 1065.004.

(b) Write down the steps of ADMM and implement for this problem (you may use CVXPY to solve each device's local optimization in each ADMM iteration). Experiment with a few values of the parameter $\rho = 1/\lambda$ from lecture to see its effect on the convergence rate of the algorithm. Plot the norm of the energy balance residual, versus iteration. Plot the power profiles of the generator and battery, as well as the energy balance residual, for several values of iteration (say, after one iteration, after 10 iterations, and after 50). Check the results against the solution found by CVXPY.

2. *The shortest LP solver.* Consider the linear program

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b \\ & x \geq 0, \end{array}$$

with $c \in \mathbf{R}^n, A \in \mathbf{R}^{m \times n}, b \in \mathbf{R}^m$. In this problem, we will consider the shortest known LP solver. For a history of short LP solves and a few other examples, we refer the

interested reader to Jacob Mattingly's collection. The LP solver we consider is adapted from one developed by Borja Peleato.

(a) Show that the following (Julia) code implements ADMM for the LP above, starting from any initial iterates x and $z \in \mathbf{R}^n$.

```
for i=1:99;z=max(z-x,x-c);x=z+A\(b-A*z);end
```

Derive the ADMM steps.

(b) Implement this LP solver using Python, and try it on a few randomly generated LPs. Generate random LPs in the same way as you did for Homework 4 Problem 3.3 with $m = 10$ and $n = 20$. This will give you an optimal $x^\star$, but it may not be unique. Run your algorithm for 1000 iterations with $\rho = 1$. Plot the objective difference, $c^T x^k - c^T x^\star$, over the iterations. Additionally, show that your solution in the last iteration is feasible.

(c) Extra credit: find an LP solver whose code is fewer than 44 character (in your language of choice).

3. *Computer-assisted analysis of the gradient method.* Consider the problem of analyzing the gradient method for solving

$$f^\star = \min_{x \in \mathbf{R}^d} f(x), \tag{1}$$

where f is L-smooth and μ-strongly convex, and where $x^\star = \operatorname{argmin}_x f(x)$ and $f^\star = f(x^\star)$ its optimal value. We use gradient descent of the form

$$x^{k+1} = x^k - \gamma_k \nabla f(x^k),$$

with a pre-determined sequence of stepsizes $\{\gamma_k\}_k$.

Our goal is to compute the smallest convergence rate $\beta(\mu, L, \gamma_k)$, such that the inequality

$$\|x^{k+1} - x^\star\|^2 \leqslant \beta(\mu, L, \gamma_k)\|x^k - x^\star\|^2, \tag{2}$$

is valid for any dimension d, for any L-smooth μ-strongly convex function f (*i.e.*, $f \in \mathcal{F}_{\mu,L}$) and for all $x^k, x^{k+1} \in \mathbf{R}^d$. This can be obtained by maximizing the value of the left-hand side of the inequality subject to $\|x^k - x^\star\|^2 \leq 1$, $x^{k+1} = x^k - \gamma_k \nabla f(x^k)$, and $x^\star \in \operatorname{argmin}_x f(x)$.

(a) Assuming $x^\star \neq x^k$ (without loss of generality), show that

$$\begin{aligned} \beta(\mu, L, \gamma_k) = \sup_{\substack{d, f \\ x^k, x^{k+1}, x^\star}} \quad & \|x^{k+1} - x^\star\|^2 \\ \text{s.t.} \quad & f \in \mathcal{F}_{\mu,L} \\ & x^{k+1} = x^k - \gamma_k \nabla f(x^k) \\ & \nabla f(x^\star) = 0 \\ & \|x^k - x^\star\|^2 \leq 1, \end{aligned} \tag{3}$$

where f, x^k, x^{k+1}, $x^\star$, and d are the variables of the optimization problem.

(b) Write down an equivalent formulation of (3) using x_{k+1} and interpolation points $\{x^i, g^i, f^i\}_{i=k,\star}$, where $g^i = \nabla f(x^i)$ and $f^i = f(x^i)$.

(c) Using the change of variables

$$G = \begin{bmatrix} \|x^k - x^\star\|^2 & (g^k)^T(x^k - x^\star) \\ (g^k)^T(x^k - x^\star) & \|g^k\|^2 \end{bmatrix}, \quad F = f^k - f^\star, \tag{4}$$

(note that $G = [x^k - x^\star \quad g^k]^T[x^k - x^\star \quad g^k] \succcurlyeq 0$), show that $\beta(\mu, L, \gamma_k)$ can be computed as

$$\begin{aligned} \sup_{G,F} \quad & G_{1,1} + \gamma_k^2 G_{2,2} - 2\gamma_k G_{1,2} \\ \text{s.t.} \quad & F + \tfrac{L\mu}{2(L-\mu)} G_{1,1} + \tfrac{1}{2(L-\mu)} G_{2,2} - \tfrac{L}{L-\mu} G_{1,2} \leqslant 0 \\ & -F + \tfrac{L\mu}{2(L-\mu)} G_{1,1} + \tfrac{1}{2(L-\mu)} G_{2,2} - \tfrac{\mu}{L-\mu} G_{1,2} \leqslant 0 \\ & G_{1,1} \leq 1 \\ & G \succcurlyeq 0. \end{aligned} \tag{5}$$

You need to show that your formulation in part (b) and (5) are equivalent. In other words, you need to construct a feasible point of one problem from a feasible point of the other problem with the same objective value and vice-versa.

(d) Implement (5) in both CVXPY and PEP to compute $\beta(\mu, L, \gamma_k)$ for different μ chosen from $\{0.1, 0.2, \cdots, 0.9\}$, $L = 1$ and $\gamma_k = \frac{2}{\mu+L}$.

- Show that the CVXPY and the PEP worst-case objective values are the same.
- Recall the strongly convex and smooth convergence result for gradient descent in previous lectures. Plot these theoretical convergence rates along with $\beta(\mu, L, \gamma_k)$ generated by CVXPY and PEP in the same figure (let μ be the x-axis of your plot). What can you comment on your plot?

Hint: use the quick start guide of PEP.